Appendix A

Guide for Nexmon firmware patching on Pi 3B+ and 4B and Experiment setup

A.1 Instructions to path the firmware

1. Download the Raspberry Pi official operating system (OS) version . Raspbian Buster Kernel v4.19.97 to image the Raspberry Pi.

2. Install following packages [39] [37]:

   (a) apt-get update **"NO upgrade-** it will load the Wi-Fi kernel and OS with the latest Kernel."

   (b) sudo apt-get install texinfo

   (c) sudo apt-get install autoconf

   (d) sudo apt-get install libtool

   (e) sudo apt-get install tcpdump

   (f) sudo apt-get install xrdp

   (g) sudo apt-get install aircrack-ng

A.1.1 Autoinstall

(a) **To install with bash follow as:** [39] sudo apt update
sudo apt install git libgmp3-dev gawk qpdf bc bison flex libssl-dev make automake texinfo libtool-bin tcpdump tmux libncurses5-dev

sudo reboot

(b) Get Kernel Headers

sudo su

sudo wget https://raw.githubusercontent.com/RPi-Distro/rpi-source/master/rpi-source -O
/usr/local/bin/rpi-source sudo chmod +x /usr/local/bin/rpi-source
/usr/local/bin/rpi-source -q –tag-update
rpi-source

sudo reboot

(c) Install Nexmon_CSI
sudo su

wget https://raw.githubusercontent.com/zeroby0/nexmon_csi/release-pi-buster-4.19.97-plus/releases/pi3Bplus-buster-4.19.97/install.sh -O install.sh
tmux new -c /home/pi -s nexmon 'bash install.sh | tee output.log'

A.2 Manual install

(a) Check the default installed kernel and default firmware version by commands (cmd)
Wifi kernel with following cmd
lshw iw dev
wireless available channels and bandwidth along with power

iw phy0 info
The output should be something like
phy#0 Interface mon0
ifindex 4
wdev 0x3
addr xx:xx........:xx
type monitor
channel 36 (5180 MHz), width: 20 MHz, center1: 5180 MHz
txpower 31.00 dBm
Unnamed/non-netdev interface
wdev 0x2
addr xx:xx:xx.....:xx6
type P2P-device
txpower 31.00 dBm
Interface wlan0
ifindex 3
wdev 0x1
addr dc:a6:32:a2:ca:d6 type managed
channel 36 (5180 MHz), width: 20 MHz, center1: 5180 MHz
txpower 31.00 dBm
Supported interface modes:

* IBSS
* managed
* AP
* monitor
* P2P-client
* P2P-GO
* P2P-device
....

OS kernel with
uname -a

A.2.1 steps for firmware patch installation

1. Install the kernel headers to build the driver and some dependencies:
   sudo apt install raspberrypi-kernel-headers git libgmp3-dev gawk qpdf bison flex make

2. Clone our repository:
   git clone https://github.com/seemoo-lab/nexmon.git

3. Go into the root directory of our repository:
   cd nexmon

4. Check if /usr/lib/arm-linux-gnueabihf/libisl.so.10 exists, if not, compile it from source:

   cd buildtools/isl-0.10
   ./configure
   make
   make install
   ln -s /usr/local/lib/libisl.so /usr/lib/arm-linux-gnueabihf/libisl.so.10

5. Check if /usr/lib/arm-linux-gnueabihf/libmpfr.so.4 exists, if not,
   compile it from source:
   cd buildtools/mpfr-3.1.4
   ./configure
   autoreconf -f -i
   make
   make install
   ln -s /usr/local/lib/libmpfr.so /usr/lib/arm-linux-gnueabihf/libmpfr.so.4

6. Then, you can set up the build environment for compiling firmware patches
   Setup the build environment:
   cd nexmon
   source setup_env.sh
   Compile some build tools and extract the ucode and flashpatches
   from the original firmware files:
   make

7. Go to the patches folder for the
   bcm43430a1/bcm43455c0 chipset: for 7_45_189>/nexmon/
   cd patches/bcm43455c0/7_45_189/nexmon Compile a patched firmware:

   make

   Generate a backup of your original firmware file:
   make backup-firmware

   copy backup files

   cp /lib/firmware/brcm/brcmfmac43455-sdio.bin brcmfmac43455-sdio.bin.orig
   Install the patched firmware on your RPI

   make install-firmware

8. To load the patched firmware on bootload run the following commands

   For 7_45_189

   cp /home/pi/nexmon/patches/bcm43455c0/7_45_189/nexmon_csi/
   brcmfmac_5.4.y-nexmon/brcmfmac.ko /home/pi

   cp brcmfmac.ko /lib/modules/5.4.79-v7l+/kernel/drivers/
   net/wireless/broadcom/brcm80211/brcmfmac

9. Install nexutil: from the root directory of our repository
   switch to the nexutil folder:
   cd utilities/nexutil/ Compile and install nexutil:

   make
   make install

   remove wpa_supplicant for better control over the WiFi interface:

   apt-get remove wpasupplicant

10. check again the modified firmware as explained in 2a

11. reboot the system
    reboot

A.3 Usage

NEXMON workinga db access to the Raspberry Pi as follow:

Connect the Pi with Router using a **LAN** cable only (**Please do not enable the Wi-Fi NIC card**), Give IP in Router's domain, the already configured IP's for Raspberry Pi's are as follows

RED_192.168.0.81 (In this PI is the most stable running Nexmon Patch) —let's call it Nexmon Pi

Black_192.168.0.82
White_192.168.0.83
Yellow_192.168.0.84

Run the following command on Pi (RED)

1. Configure to tune the Channel 36/40 Hz of Raspberry PI
   mcp -c 36/40 -C 1 -N 1

2. You will get this output:

   JtgBEQAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
   AAAAAAA==

3. run further following commands
   ifconfig wlan0 up

   nexutil -Iwlan0 -s500 -b -l34 -vJtgBEQAAAAAAAAAAAAAAA
   AAAAAAAAAAAAAAAAAAAAAAAAA==

   iw dev wlan0 interface add mon0 type monitor

   ip link set mon0 up

4. To dump the CSI data

   tcpdump -i wlan0 dst port 5500 -w <Filename_40MHz_001>.pcap

   It will start dumping the CSI data in the Home directory of Pi

Now connect any other Pi or device to create some traffic on the router to utilize the 40 MHz bandwidth. If there is low traffic on the router, it may go to power management mode or switch to a lower frequency band.It may further detune the Nexmon Pi to a lower frequency band.

The Pi can be configured for 20 MHz and 80 MHz frequency as well
Run the following commands accordingly and follow steps 2 to 4, the output base 64 code would be different for different frequency bands.

For 20 MHz
mcp -c 7/20 -C 1 -N 1

For 80 MHz
mcp -c 36/80 -C 1 -N 1

where **c** is the channel given by command c/frequany

A.3.1 Parallel monitoring to ensure if the CSI is getting dumped

Run Wireshark as root on Pi

Select the wireless card in Wireshark and monitor the traffic on udp port 5500 It should show the running traffic as the following figure.

Note: running the Wireshark for more extended time on Pi with a graphical interface may hang the Pi because of limited resources.

A.3.2 Splitting and trimming the collected CSI data

Splitting and trimming the collected CSI data of .pcap files

After collecting the data, if it is required to trim the few packets from the beganing and at the end of the file, use the following commands.

editcap -r <filenameToProcess> <filenameToSave> Packet start number - Packet end number

Save data file with 150 packets in each file for different activities

editcap -c 300 -F pcap

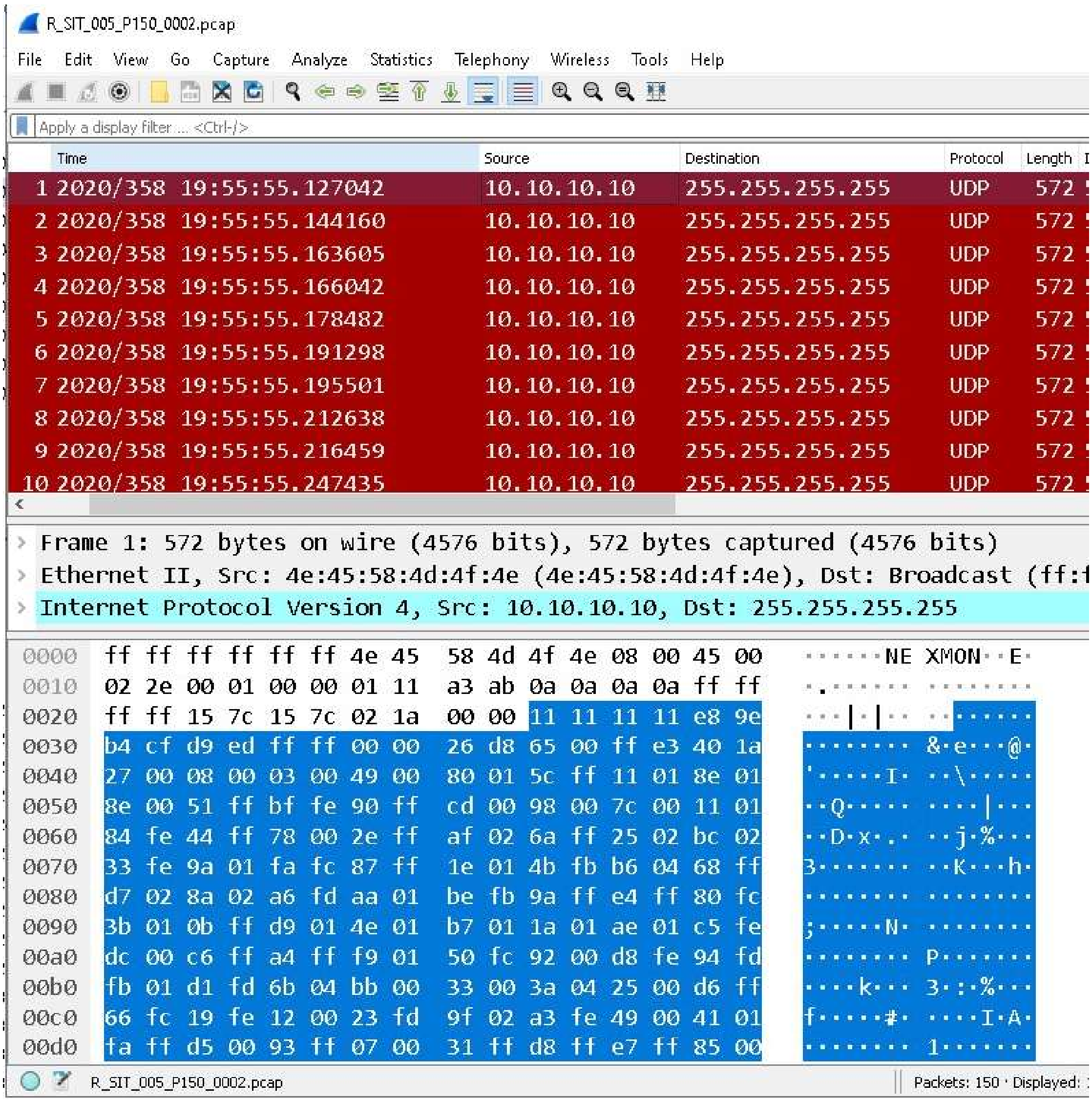

Figure A.1: Wireshark

A.4 MATLAB code execution guide

Execute the $MainFile\_SVM\_LSTM\_NB\_Rd\_Final1.m$ file; It will run SVM, Naive Bayes, LSTM on $CSI_{feature}$ data *and* on $CSI_{denoised}$ *data*.

If there is new data to be tested, please change the following Matlab code files database path.

Change the folder name in $MainFile\_SVM\_LSTM\_NB\_Rd\_Final1.m$, $SVM\_Data.m$ and $SVM\_Test.m$

For example, use the template folder for the new database and change the name Template_ to Dataxx for all entries. There is a separate folder for each type of activity. The Code automatically divide the files for Training, Test in 80:20 ratio, and Validation with 10%. Folder structure is:

```
\Train\Dataxx_Database\room\Red\EMPTY

\Train\Dataxx_Database\room\Red\SIT

\Train\Dataxx_Database\room\Red\STAND

\Train\Dataxx_Database\room\Red\WALK
```

Figure A.2: Database folder structure